1. Define "soft real-time". List 3 examples of soft real-time requirements.

- soft real-time is a system that runs in real-time that has deadlines for tasks, However missing the deadline for tasks is not detrimental to the system. i.e. the system can still run without damage to user or the system itself.

examples
- smart watch
- smart fridge
- Car radio

2. Define "hard real-time". List 3 examples of hard real-time requirements.

- Hard real-time is similar to soft real-time such that it has deadlines for tasks, However missing the dealines is dangerous and potentially fatal. Missing deadlines can lead to loss of life or severe damage to system(financial penalty).

examples
- Radar's
- missile guidance
- drone control

3. What is the purpose of a real-time kernel such as FreeRTOS?

- It allows the user to program with the capabilities of an OS that is made specifically for real-time operation. This allows the user to better manage timing constraints and allows for the possibility for multi-threading for tasks.

4. List 2 pros and 2 cons of using free software such as FreeRTOS.

pros
- Free to use for the public
- Can be used to make commercial products

cons
- Lack of official support from a company
- potential lack of features from a paid license products

5. List 5 benefits of using an RTOS.
- Very flexible with task priority assignments
- Queues
- Event groups
- Can be Pre-emptive
- Use's software(easier turn around then HDL's)